TRANSFUSION MEDICINE
Official Journal of the British Blood Transfusion Society
British Blood Transfusion Society
Transfusion Medicine | ORIGINAL ARTICLE

Improving the screening of blood donors with syphilis rapid diagnostic test (RDT) and rapid plasma reagin (RPR) in low- and middle-income countries (LMIC)

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Received 22 April 2016; accepted for publication 14 September 2016

SUMMARY

Background: Syphilis testing conventionally relies on a combination of non-treponemal and treponemal tests. The primary objective of this study was to describe the positive predictive value (PPV) of a screening algorithm in a combination of a treponemal rapid diagnostic test (RDT) and rapid plasma reagin (RPR) test at Komfo Anokye Teaching Hospital (KATH), Ghana.

Materials and Methods: From February 2014 to January 2015, 5 mL of venous blood samples were taken from 16 016 blood donors and tested with a treponemal RDT; 5 mL of venous blood was taken from 526 consenting initial syphilis sero-reactive blood donors. These RDT reactive samples were confirmed with an algorithm, applying the Vitros®/Abbott-Architect® algorithm as gold standard.

Results: A total of 478 of 526 RDT reactive donors were confirmed positive for syphilis, making a PPV of 90·9%. Of the 172 (32·7%) donors who were also RPR positive, 167 were confirmed, resulting in a PPV of 97·1%. The PPV of the combined RDT and RPR (suspected active syphilis) testing algorithm was highest among donors at an enhanced risk of syphilis, family/replacement donors (99·9%), and among voluntary donors above 25 years (98·6%).

Discussion: Screening of blood donors by combining syphilis RDT and RPR with relatively good PPV may provide a reasonable technology for LMIC that has a limited capacity for testing and can contribute to the improvement of blood safety with a minimal loss of donors.

Key words: blood donors, blood safety, public health, rapid diagnostic test, rapid plasma reagin.

Syphilis screening is a big challenge in many low- and middle-income countries (LMIC) that have a limited capacity for testing. High syphilis prevalence among healthy blood donors in Africa aggravates the problem in this region. Techniques for syphilis testing are very problematic and conventionally rely on a combination of non-treponemal and treponemal tests. The non-treponemal antibody tests for donor screening include the Venereal Disease Research Laboratory (VDRL) (Harris *et al.*, 1948) and the rapid plasma reagin (RPR) (Larsen *et al.*, 1998). The advantages are that these tests are inexpensive, fast, simple to perform and more sensitive (Montoya *et al.*, 2006). They are able to identify the contaminated blood donors a few days before the treponemal test and thus are useful for acute infection. Moreover, quantifiable titers can establish a baseline to evaluate treatment response (Sena *et al.*, 2010a) as they usually revert to negative after successful treatment (Larsen, 1989; Romanowski *et al.*, 1991; Larsen *et al.*, 1998). However, VDRL and RPR cannot be automated and are therefore time-consuming if used for large-scale testing. In addition, results may be difficult to interpret, and this requires the sufficient training of health personnel to ensure correct testing and interpretation. Another major problem when using non-treponemal tests is the possibility of biological false positive reactions due to cross-reactivity with molecules in other conditions, such as viral infections, pregnancy, malignant neoplasms, autoimmune diseases and advanced age (Larsen *et al.*, 1995; Sena *et al.*, 2010a).

Treponemal tests for donor screening classically included the *Treponema pallidum* haemagglutination assay (TPHA), the

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doi: 10.1111/tme.12363

T. pallidum passive particle agglutination (TP-PA) assay and the fluorescent treponemal antibody absorption (FTA-ABS) (Hunter *et al.*, 1964). However, newer, automated treponemal tests have reduced running costs and provide objective readings, making them useful for large blood centres. Treponemal tests typically remain positive even after treatment (Schroeter *et al.*, 1973; Gerber *et al.*, 1997; Larsen *et al.*, 1998), implying that a donor previously diagnosed for syphilis cannot be distinguished from a new or untreated case of syphilis.

In developing countries and areas with limited resources, laboratory facilities are often unavailable for standard automated syphilis tests. Given the barriers to automated testing, many resource-limited countries are resorting to syphilis rapid diagnostic tests (RDTs) for transfusion-transmitted infection (TTI) screening (Kaur & Kaur, 2015). Although quality RDTs have the potential to increase the safety of the region's blood supply, uncertainty surrounding the performance of some RDTs in the field has increased debate regarding their application to TTI screening (Scheiblauer *et al.*, 2006; Laperche & Francophone African Group for Research in Blood, 2013; Mbanya, 2013).

Some rapid tests are highly sensitive and specific (Owusu-Ofori *et al.*, 2005; Sena *et al.*, 2010a) but cannot differentiate between active and treated syphilis; others may give false positive reactions (van Dommelen *et al.*, 2008). Most rapid tests detect IgM, IgG and IgA antibodies and involve immunochromatographic strips in which one or multiple *T. pallidum* recombinant antigens are applied to nitrocellulose strips as a capture reagent. Irrespective of the advantages of these rapid tests, if they have a low positive predictive value (PPV) – high false positive rate – and are used in blood banks, then apparently, many donors will be deferred when they carry no risk to the blood supply. Conversely, if the PPV is high (low false positive rate), then few donors with no risk to the blood supply will be deferred. This is really important in LMIC settings where blood supplies are critical, coupled with a high burden of TTIs.

The Transfusion Medicine Unit (TMU) of the Komfo Anokye Teaching Hospital (KATH, Kumasi, Ghana) has been practicing pre-donation screening with viral RDTs for blood donors since 2000 (Owusu-Ofori *et al.*, 2005; Allain *et al.*, 2010). KATH operates on two types of blood donors, namely, voluntary non-remunerated blood donors (VNRBD), constituting 70% of blood collection, and family replacement donors (FRD), constituting 30% of blood collection for over a decade now. The decision to introduce syphilis screening as part of pre-donation screening was decided by the Hospital Transfusion Committee in 2012. However, a loss of blood for transfusion of 7% by syphilis RDT reactive donors was a threat to blood supply, and an algorithm that combined robustness, safety and minimal discard rates was needed. KATH implemented an algorithm consisting of syphilis testing by RDT before blood collection followed by RPR (IMMUTREP RPR, Omega Diagnostics – Scotland, UK) testing of syphilis RDT reactive donors (Sarkodie *et al.*, 2016a,2016b). The presumption was that by discarding only .donors who were also positive for RPR, only donors at risk of active syphilis were rejected, whereas donors with previous well-treated syphilis could continue to donate, and their blood would be released for transfusion. One of the risks is that although combining two different syphilis tests, the algorithm might still defer too many donors due to false reactivity in the two tests. In that case, the algorithm would still compromise blood supply, and it would expose donors to unneeded worries, stigma and therapy.

In order to study this aspect, the present study validates the PPV of the implemented algorithm by applying a gold standard retest algorithm combining two different automated anti-TP immune assays of 526 syphilis RDT reactive donors in improving blood donor screening.

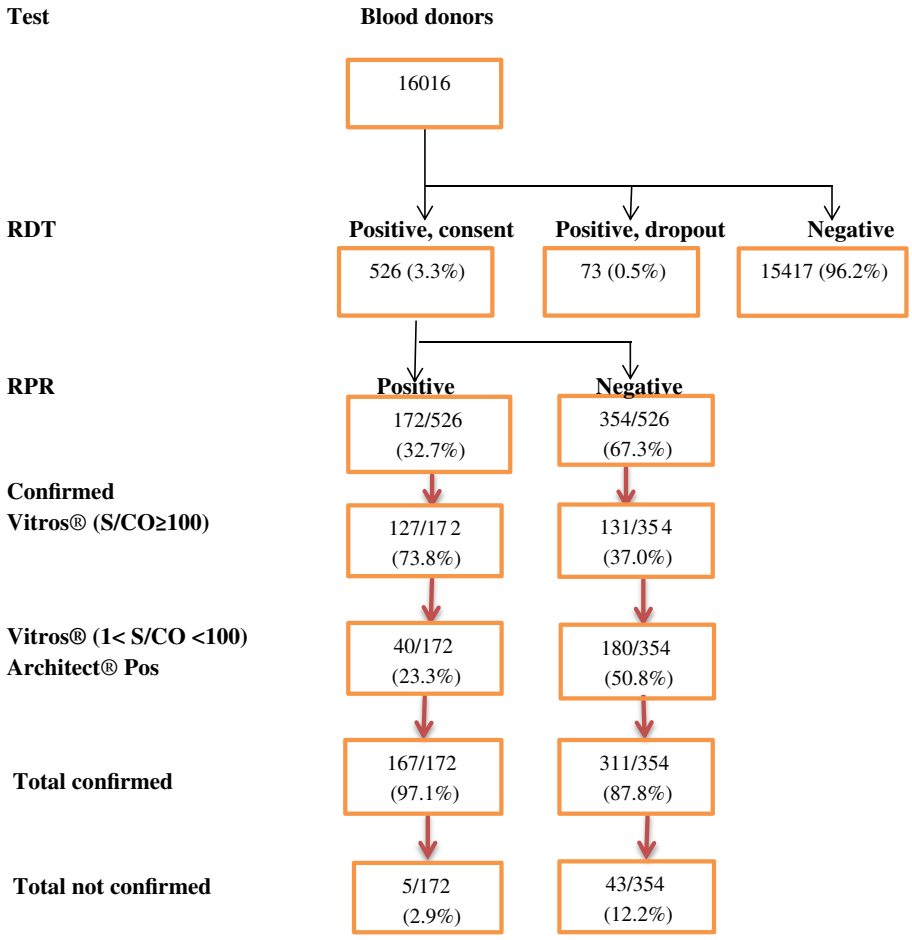
MATERIALS AND METHODS

We conducted a descriptive cross-sectional study between February 2014 and January 2015. A total of 16 016 blood donors were initially tested according to routine standard operational procedures with a treponemal RDT (Fortress® Diagnostics Limited – Antrim, UK). Sensitivity and specificity of the Fortress RDT are stated to be 99·7 and 99·6%, respectively, for the qualitative detection of antibodies (IgG and IgM) to *T. pallidum* in serum or plasma. A total of 5 mL of venous blood was taken from 526 consenting initial Fortress RDT syphilis sero-reactive blood donors. All these samples were further tested according to routine standard procedures with RPR (BD Macro-Vue™ Card test, Branchburg, New Jersey, USA) to identify potential active syphilis infections.

For gold standard confirmation of the Fortress RDT, all 526 RDT reactive samples were subsequently retested in an algorithm combining two automated treponemal immunoassays and a treponemal immunoblot. Initial retesting was performed by the Vitros® Syphilis *Treponema Pallidum* Antibody (TPA) chemiluminescence immunoassay using the Vitros ECi/ECiQ Immunodiagnostic Systems described elsewhere (Gonzalez *et al.*, 2015). Briefly, the Vitros Syphilis TPA assay is a qualitative assay that detects total antibodies (IgG and IgM) to *Treponema pallidum* (TP) reacting with biotinylated and horseradish peroxidase (HRP)-labelled recombinant TP antigens TP15, TP17, TP47 and is bound to streptavidin-coated wells. The illuminating reaction detected from the bound HRP conjugates is directly proportional to the concentration of anti-TP antibodies, and high signal samples [signal at Cutoff (S/CO) >100] were considered confirmed (Fig. 1). The assay was mainly validated in a western population (data not shown), with a specificity of 99·8% (CI 98·7–100%) and a sensitivity of 100% using Syphilis Mixed Titer Performance Panel PSS202 (BBI Diagnostics, Bridgend, UK) and clinical samples from known syphilis-treated patients of both Caucasian and African origin.

All Vitros low reactive samples (S/CO <100) were additionally tested with another qualitative anti-TP immunoassay Architect® Syphilis TP (Abbott Diagnostics, Abbott Park, IL, USA), which also detects antibodies binding to the recombinant TP antigens TpN15, TpN17 and TpN47 (Fig. 1). A reactive

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RDT; - rapid diagnostic test, RPR; - rapid plasma reagin, S/CO-; sample over cut-off
N/B-Architect® was performed on specimen that were low reactive with Vitros® (1< S/CO <100)

Fig. 1. Algorithm for syphilis confirmatory testing of sero-reactive blood donors, Transfusion Medicine Unit, Komfo Anokye Teaching Hospital.

Fortress RDT sample was considered confirmed positive for specific anti-TP antibodies if the Vitros Syphilis TPA was highly reactive (S/CO > 100) or if Vitros Syphilis TPA was low reactive (1 < S/CO <100) and Architect Syphilis TP reactive.

As a quality control measure, 78 of 526 syphilis RDT sero-reactive samples were further tested in a line immunoassay (LIA) (Furijebio, Ghent, Belgium) (Fig. 2). The LIA detects individuals binding to the same recombinant TP antigens TpN15, TpN17 as well as to TpN47 and a synthetic peptide TmpA derived from *T. pallidum* proteins (Ebel *et al.*, 2000).

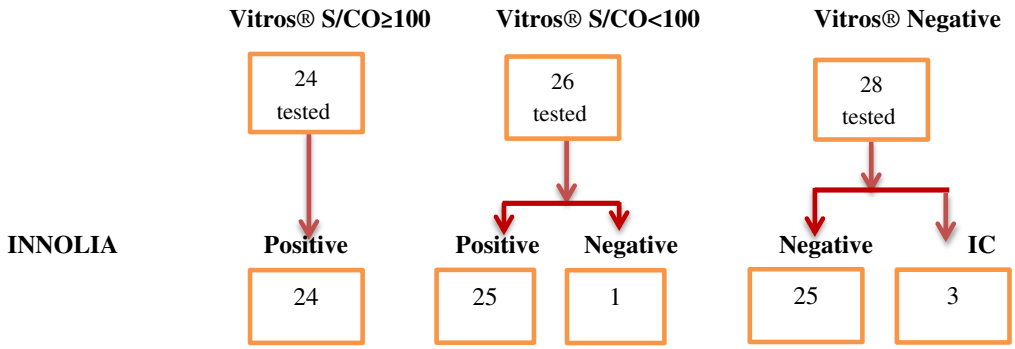
Syphilis sero-reactive donors were informed of the study and signed an informed consent form according to the study protocol, which was approved by the Ethics Committees of Kwame Nkrumah University of Science and Technology (KNUST) Kumasi, Ghana (CHRPE/AP/423/13) and Liverpool School of Tropical Medicine, UK (18/02/2014).

Background data were recorded onto a spreadsheet consisting of gender, age, number of donations, donor type and routine testing results. Data were then exported to STATA (STATACORP, Texas, version 12·0) for analysis. We estimated PPVs by calculating proportions and providing their respective confidence intervals. Multivariable logistic regression was performed on syphilis reactivity as an outcome. Age, gender and donor type were included as independent variables, with results presented as odd ratios and 95% confidence intervals. A *P*-value of <0·05 denoted a statistically significant difference in all statistical comparisons.

RESULTS

A total of 599 out of 16 016 blood donors reacted to the syphilis Fortress test, making an estimated sero-prevalence of 3·7% (95% CI 3·5–4·1%). A total of 73 (12·2%) blood donors who reacted with the Fortress syphilis test were excluded from the study (Fig. 1) because 41 (6·8%) of them were co-infected with hepatitis B virus (HBV), 15 (2·8%) were co-infected with HIV and 7 (1·2%) were co-infected with hepatitis B virus (HCV), while 10 (1·7%) did not consent. Thus, 526 (3·3%) syphilis sero-reactive blood donors were included in the study, of whom 199 (37·8%)

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Approximately 15 % of the samples tested with Vitros® were further tested with INNO-LIA as a quality control of the validation algorithm.
IC; - Inconclusive, S/CO-; sample over cut-off

Fig. 2. Quality control with INNOLIA for syphilis testing of sero-reactive blood donors, Transfusion Medicine Unit, Komfo Anokye Teaching Hospital.

Table 1. Syphilis RDT sero-reactive blood donors

	RPR		
RDT-reactive blood donors	Positive (%)	Negative (%)	Total (%)
Confirmed	167 (97·1)	311 (87·8)	478 (90·9)
Unconfirmed	5 (2·9)	43 (12·2)	48 (9·1)
Total	172 (32·7)	354 (67·3)	526

were VNRBD (95% CI 33·8–42·1%). Generally, blood donors tested were aged 16–59 years, with a mean age of 25 (SD = 9·1), compared with syphilis sero-reactive donors, who showed an age range of 17–53 years, with a mean age of 31 years ($P < 0·001$).

PPV of syphilis RDT reactive samples according to donor type

Of the 526 RDT syphilis sero-reactive samples tested with Vitros, 478 were reactive and confirmed by the algorithm, making a PPV of 90·9% (Table 1). Similarly, the proportion or PPV of sero-reactive FRD (309/327, 94·5%) that was confirmed (Table 2) was statistically significantly higher than that of confirmed sero-reactive VNRBD (169/199, 84·9%) ($P = 0·001$), although there was 100% PPV in some age groups. Of the total blood donors tested, 10 218 (63·8%) were VNRBD, of whom 199 (1·95%) were syphilis sero-reactive (95% CI 1·73–2·28%) as shown in Table 2. Of the 5798 FRD tested, 327 were syphilis sero-reactive, which was significantly higher [5·64% (95% CI, 5·08–6·26%)] $P < 0·001$ compared with VNRBD.

PPV of syphilis RDT reactive samples according to donor age

Syphilis RDT sero-reactive donors showed a range of 17–53 years, with a mean age of 31 years. The PPV of the VNRBD ranges from 74·4 to 100% for the ages ranging between 17 and 55 years, whilst that of FRD ranges from 91·2 to 100% for the same age difference (Table 2). Generally, the PPVs of the syphilis-confirmed reactive donors increase with age for all donor types except in VNRBD, where those aged between 46 and 55 years have a relatively lower prevalence (3·17%) but higher PPV (100%). Although syphilis-confirmed sero-prevalence of FRD was 5·33%, there was higher prevalence as the age increased. Similarly, the prevalence of syphilis-confirmed VNRBD was 1·65%, although there was a higher prevalence of 5·24% in the 36–45 age category.

PPV of syphilis RDT and RPR reactive samples

Of the 526 syphilis RDT reactive samples, 172 (32·7%) were RPR positive (95% CI 28·8–36·8%). Out of these, 167 were confirmed, making a PPV of 97·1% (Table 1). Thus, the PPV was higher among RDT and RPR reactives (97·1%) than in the total population of RDT reactives (90·9%) as shown in Table 1. Conversely, the PPV was higher among FRD (99·1%) than in VNRBD (93·3%). More FRD and more donors aged 26 and over were RPR positive and increased with age as the PPV increased (Table 2). Similarly, the PPV of RDT and RPR dual-reactive donors was the highest among FRD and among donors aged 26 and above (Table 2). Out of the five RPR false positives, four were VNRBD, out of which three were aged between 16 and 25 years, and one was aged between 26 and 35 years, while the other one (FRD) was between 26 and 35 years. Additionally, 311 of 354 (87·8%) RPR-negative donors tested positive with Vitros.

The effect of age, gender and donor type on syphilis reactivity

By multivariable logistic regression, we showed a positive association between syphilis reactivity and all included explanatory

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Table 2. Syphilis RDT and RPR sero-reactive blood donors confirmed with Vitros TP, stratified according to age

Blood donors	Total RDT tested (%)	RDT + (%)	RDT+, Vitros+ (%, PPV)	RDT+, RPR+ (%)	RDT+, RPR+, Vitros+ (%, PPV)
VNRBD	10218 (63·8)	199 (1·95)	169 (1·65, 84·9)	60 (0·59)	56 (0·55, 93·3)
Age group (years)					
≤25	7525 (73·6)	78 (1·04)	58 (0·77, 74·4)	27 (0·36)	24 (0·32, 88·9)
26–35	1750 (17·1)	76 (4·34)	68 (3·89, 89·5)	23 (1·31)	22 (1·26, 95·7)
36–45	687 (6·7)	38 (5·53)	36 (5·24, 94·7)	9 (1·31)	9 (1·31, 100·0)
46–55	221 (2·2)	7 (3·17)	7 (3·17, 100·0)	1 (0·30)	1 (0·30, 100·0)
>56	35 (0·4)	0	0	0	0
Gender					
Female	3477 (34·0)	24 (0·69)	16 (0·46, 66·7)	6 (0·17)	6 (0·17, 100·0)
Male	6741 (66·0)	175 (2·60)	153 (2·27, 87·4)	54 (0·80)	50 (0·74, 92·6)
FRD	5798 (36·2)	327 (5·64)	309 (5·33, 94·5)	112 (2·12)	111 (1·91, 99·1)
Age group (years)					
≤25	1869 (32·2)	57 (3·05)	52 (2·78, 91·2)	24 (1·28)	23 (1·23, 95·8)
26–35	2405 (41·5)	148 (6·15)	139 (5·78, 93·9)	48 (2·00)	48 (2·00, 100·0)
36–45	1186 (20·4)	95 (8·01)	91(7·67, 95·8)	32 (2·70)	32 (2·70, 100·0)
46–55	321 (5·5)	27 (8·41)	27 (8·41, 100·0)	8 (2·50)	8 (2·50, 100·0)
>56	18 (0·4)	0	0	0	0
Gender					
Female	538 (9·3)	13 (2·42)	13 (2·42, 100·0)	4 (0·74)	4 (0·74, 100·0)
Male	5260 (90·7)	314 (5·97)	296 (5·63, 94·1)	108 (2·05)	107 (99·1)
Total	16 016	526 (3·28)	478 (2·98, 90·9)	172 (1·07)	167 (1·04, 97·1)

+, positive.

parameters: increased age, male gender and status as a FRD (Table 3). The effects of male gender and FRD status were similar whether a positive end-point was defined as RDT+, RDT+ Vitros+, RDT+ RPR+ or RDT+ Vitros+ RPR+ (Table 3). The effect of age was weaker for end-points including RPR reactivity (Table 3). The male gender was a stronger predictor of syphilis reactivity than status as an FRD (Table 3).

Samples tested with INNO-LIA as quality control

Approximately 58% (28) of the samples that were negative according to the Vitros test were tested with INNO-LIA. A total of 25 of them were negative, while the rest (3) were inconclusive. All 24 (∼9%) samples that were Vitros-high reactives were confirmed with INNO-LIA, and 25 of 26 low with Vitros were confirmed with LIA, and one was inconclusive.

DISCUSSION

Syphilis infection in blood donors continues to pose a major threat in many developing countries, including Ghana (Adjei *et al.*, 2003; Sarkodie *et al.*, 2016a,2016b). The Fortress syphilis RDT that was used for this study based on its performance characteristics has a sensitivity and specificity of 99·7 and 99·6%, respectively, according to the manufacturer. When compared to the gold standard in single testing, it gave a PPV of 90·9%. Similar syphilis RDTs in other studies gave a PPV of 95·2%, with sensitivity and specificity of 93·6 and 92·5%, respectively (Sato *et al.*, 2003). As PPV relies on both test specification and disease prevalence, it is not surprising that other studies have shown PPVs of some RDTs to be both lower and higher than this value (Pruett *et al.*, 2015). When combining a RDT with a non-specific syphilis test, in this case RPR, a much higher PPV was achieved (97·1%). Thus, by combining the two tests, both donors with confirmed but inactive TP infections and donors with unspecific RDT reactions could avoid deferral, and they could therefore still contribute to the blood supply. The key concept underlying blood safety, especially in LMIC, is the balance between blood supply and blood safety in the context of a poor blood supply, high prevalence of TTI compared and limited resources. As stated earlier, if there is a low PPV (high false positive rate), then many donors will be deferred when they carry no risk to the blood supply. Furthermore, a screening test with a low PPV/high false positive rate has the potential to cause unnecessary harm to blood donors because of fears and wrong information.

The use of syphilis RDTs in resource-poor settings

Pre-donation screening for TTIs with syphilis RDTs is a strategy that has been proposed for use in resource-poor, high-prevalence settings without access to a stable pool of low-risk donors (Salawu & Murainah, 2006). One of the reasons behind this was to reduce blood bag wastage and associated costs of consumables in collecting blood from donors, which was not used because of positive screening tests. One study in Ghana demonstrated savings of more than $11,000 in blood bags and testing costs over a 1-year period using pre-donation screening (Owusu-Ofori *et al.*, 2005a,2005b).

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Table 3. Multiple variable logistic prediction of syphilis reactivity

RDT + 526/16016 (3·28%)	Odds ratio	*P*-values	95% CI
Age (years)	1·04	<0·001	1·03 – 1·05
Gender (male)	2·99	<0·001	2·11 – 4·04
Donor type (FRD)	1·92	<0·001	1·60 – 2·30
RDT+, Vitros®+ 478/16016 (2·98%)			
Age (years)	1·05	<0·001	1·04 – 1·06
Gender (male)	3·41	<0·001	2·29 – 5·08
Donor type (FRD)	1·75	<0·001	1·44 – 2·14
RDT+, RPR+ 172/16016 (1·07%)			
Age (years)	1·03	<0·001	1·02 – 1·05
Gender (male)	3·22	<0·001	1·67 – 6·22
Donor type (FRD)	2·04	<0·001	1·45 – 2·88
RDT+, RPR+, Vitros®+ 167/16016 (1·04%)			
Age (years)	1·03	<0·001	1·02 – 1·05
Gender (male)	2·72	<0·001	1·49 – 4·99
Donor type (FRD)	2·06	<0·001	1·46 – 2·89

Syphilis prevalence in Kumasi blood donors

In this study, we found the prevalence of syphilis in Kumasi blood donor population with the use of RDT to be 3·7%, which is not different from previous studies in Ghana (Adjei *et al.*, 2003, 2006; Owusu-Ofori *et al.*, 2011). Like other studies, we found a higher rate of syphilis reactivity among FRD than among VNRBD. This was only partly explained through higher age and more males among FRD as FRD status was an independent positive predictor of syphilis reactivity in a logistic regression analysis. There is an ongoing struggle to have 100% VNRBD in Ghana and elsewhere in Africa, which, if successful, may reduce syphilis sero-reactivity. Despite this, family donations remain dominant on the African continent. The association between age and syphilis reactivity is most likely caused by a longer period of sexual exposure. However, a cohort phenomenon with older donors being more exposed to yaws in childhood may also contribute. However, the data confirm that younger first-time donors are safer than older donors, whereas the highest safety both with regards to infection risk and blood supply lies in a system of repeat donations as the main source of blood for transfusion (Allain *et al.*, 2010).

Syphilis sero-reactivity and active syphilis in blood donors

Our data suggest that a total of 167 or 1% of tested blood donors were confirmed syphilis RDT and RPR reactive. Our logistic regression data additionally indicate that there are independent effects of age, male gender and FRD donor type on syphilis sero-reactivity. When considering RPR reactivity, the effect of age was smaller, indicating that higher age is a stronger prediction of previous syphilis infections than of recent infections. In our previous published article (Sarkodie *et al.*, 2016a,2016b), the decrease in reactive samples from RDT to RPR is approximately six times compared to this study, which is only approximately three times. This considerable discrepancy is probably due to changes in test kits. The RPR test kit used in the previous article (IMMUTREP RPR, Omega Diagnostics – Scotland, UK) differs in sensitivity and specificity from the one used in this study (BD Macro-VueTM Card test). Additionally, testing errors on the part of the laboratory scientists in both testing procedures could contribute to the discrepancy. As a lot of blood is transfused without storage, this may constitute a significant risk of syphilis transmission through transfusion as previously reported syphilis (Owusu-Ofori *et al.*, 2011). Our data thus support the combined use of RDT and RPR to detect active syphilis of potential blood donors, which would enable more focused deferral of potential active syphilis cases for treatment. These cases of suspected active syphilis can be identified with a minimal loss of donors. It is important to repeat this study in other resource-poor settings where syphilis prevalence is high.

Strengths and limitations

The strength of this study is the description of a real-life performance with regards to the PPV of a newly suggested combined syphilis testing algorithm, combining an RDT and RPR for the identification of potential active syphilis. The algorithm used for gold standard confirmation was robust as it involved three different Treponema-specific tests used sequentially to confirm weak and negative results. The INNO-LIA assay has been shown to provide highly reliable confirmatory diagnostic information of anti-TP antibodies (Ebel *et al.*, 2000) and was furthermore used as a quality control for the confirmation algorithm of anti-TP antibodies.

Three major limitations need to be mentioned. Firstly, the assumption that RDT positive donors testing negative in RPR were without significant risk for transfusion was neither tested by recipient look back or by molecular testing of donors. Secondly, the proportion of truly syphilis-reactive donors missed by the initial RDT was not evaluated because of resource constraints.

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Finally, we cannot assume infectivity among all confirmed RPR reactive donors.

CONCLUSION

In a blood bank system like the one in Kumasi, Ghana, with a relatively high prevalence of syphilis and where infrastructure to support formal laboratory testing is often lacking, syphilis screening with RDTs may provide a reasonable technology. The combination of both RDT and RPR reduces the loss of donors and blood for transfusion. The combined RDT and RPR testing has a satisfactory high PPV, meaning that unneeded loss of blood for transfusion and false syphilis diagnoses of donors are minimised. The high PPV of a combined RDT and RPR algorithm suggests that further routine confirmation of a donor deferred with dual RDT and RPR reactivity is not needed. This adds to the robustness and cost efficiency of the suggested TP screening algorithm.

ACKNOWLEDGMENTS

We thank Prof. Daniel Ansong and Mr. Bernard Arhin, all of Research and Development Unit, Messrs. Derek Agyeman Prempeh and Eliot Eli Dogbe and the entire staff of Transfusion Medicine all of Komfo Anokye Teaching Hospital, Kumasi, not forgetting the entire T-REC team for their support. The study was funded by the European Union Seventh Framework Programme (FP7/2007-2013) under Grant Agreement No. 266194 through T-REC, a transfusion research capacity for building research in Africa, and part of PhD requirements at the Liverpool School of Tropical Medicine, UK.

Francis Sarkodie designed the study, performed the research, analysed the data and wrote the manuscript. Oliver Hassall contributed to the design, data analysis and manuscript writing. Ellis Owusu-Dabo contributed to the design, data analysis and manuscript writing. Shirley Owusu-Ofori contributed to the design and manuscript writing. Imelda Bates contributed to the design, data analysis and manuscript writing. Ib C. Bygbjerg contributed to the design, data analysis and manuscript writing. Alex Owusu-Ofori contributed to the design and manuscript writing. Lene Holm Harritshøj contributed to the confirmatory testing, analysis and manuscript writing. Henrik Ullum contributed to the design, confirmatory testing, analysis and manuscript writing.

CONFLICT OF INTEREST

The authors have no competing interests.

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